Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Tomato

1.1 Flowering and Fruit Setting

- **Flowering stages and development:** Tomato (*Solanum lycopersicum* L.) plants undergo floral bud initiation and inflorescence formation, producing hermaphroditic flowers arranged in terminal, cymose, or axillary clusters. Floral development progresses through flower opening and anthesis, featuring a differentiated perianth with five sepals and five petals. Following successful fertilization, the gynoecium transitions from ovary expansion into early fruit set. Hybrids possessing jointless pedicels (*j2*), such as Arka Apeksha and Arka Vishesh, maintain mature fruits intact on the vine even 110 days after transplanting in the main field.

- **Environmental requirements:** Tomato flowering and reproductive development require a warm and humid climate or warm conditions with an optimal temperature range of 25–27
(°C
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- **Flowering time and duration:** Documented crop-stage timelines in Maharashtra establish Kharif sowing in May–June or June second week with transplanting in June–July or July–August; Rabi sowing in September–October with transplanting in October–November; and Summer sowing in December–January or January second week with transplanting in January–February or February last week. Seedlings developing 5–6 true leaves reach transplanting stage within 4 weeks of sowing. Total crop durations span 150 days for Phule Kesari, 165 days for Dhanshree, Bhagyashree, and Phule Jayshree, 175 days, and 180–190 days for Phule Raja and related cultivars.

- **Flower morphology and reproductive biology:** Flowers of tomato (*Lycopersicon esculentum* L.) are hermaphroditic and actinomorphic, borne solitary or in cymose inflorescences. The differentiated perianth contains five sepals and five petals, the androecium features five epipetalous stamens present in multiples of four or five (most commonly four or eight) with a hypogynous disk, and the gynoecium consists of two carpels forming a superior ovary.

- **Pollination and fertilization:** Specific pollen production quantities, pollen viability thresholds, pollen germination rates, stigma receptivity periods, and pollinator vibration mechanisms are Not specified in the provided sources. Inadequate pollination or fertilization under high or low temperature extremes prevents normal seed set, causing fruit puffiness.

- **Fruit setting and initial development:** Fruit set represents the transition of the fertilized superior ovary into rapid cell division and early fruit growth. Standard commercial harvesting occurs when approximately 10% of the fruits on the first hand reach the breaker stage of maturity. Mature green fruits can be stored up to 30 days at 10 (°C), $whereas vine-ripened fruits retain quality for approximately one week.$

- **Environmental stresses and reproductive failure:** High temperature stress where maximum temperatures exceed 40.0 (°C)(($T_{max} > 40.0$°C))$directly causes heavy flower drop and fruit abortion. Temperature extremes during fruit set impair pollin$ $facing deformity at the blossom end. Persistent wet weather and high relative humidity encourage late bligh$

- **Water management:** Flowering and fruit development are critical growth stages during which water stress must not be imposed. Light irrigation is applied 3–4 days after transplanting, followed by scheduled irrigation intervals of 7–8 days during Kharif, 10–12 days during Rabi, and 5–6 days during Summer. Under high temperature conditions (($T_{max} > 40.0$°C))$, sufficient irrigation must be provided to minimize flower and fruit drop, while heavy irrigation follow$

- **Nutrient management:** Recommended fertilizer schedules for varieties require 40:24:24 kg N:P:K/acre, with 50% N applied basal before transplanting, a second split top-dressed, and a third N dose of 13.5 kg N/acre applied at 60 days after transplanting. For hybrids, recommended rates are 60:36:36 kg N:P:K/acre, with the third N dose of 20 kg N/acre applied at 60 days after planting. An alternative schedule applies N and half P and K at transplanting, 60 kg N with remaining P and K at 30 days, and 60 kg N at 50 days after transplanting. Localized calcium deficiency causes blossom-end rot, corrected by soil calcium applications and balanced water management. Boron deficiency causes water transport disruption and is corrected by foliar spray of borax @ 0.2%, copper deficiency by 0.5% $CuSO$ ($_4$)$twice fortnightly, and iron deficiency by 0.5\% FeSO$ ($_4$).

- **Plant growth regulation:** Gibberellic acid ((GA_3))$spray @ 0.001\% L (70.8 ml in 180 - -200 L of water/acre) applied at 45 and 65 days after sowing enhances pla$ $-20 mg/L increase yield at low temperatures, while seed treatment @ 40 - -100 mg/L improves seed germin$ $broadcast 2 - -3 days before sowing, or Triacontanol 0.05\% EC/0.1\% EW @ 100 ml in 160 -$ $-200 L of water/acre is sprayed at 25, 45, and 65 days after sowing. Ethephon foliar spray @ 100-$ $-500 mg/L enhances flowering, fruiting, and yield, whereas pre-harvest spray @ 1000 mg/L or post-$ $harvest dipping in ethephon 39\% SL @ 2500 ppm promotes uniform fruit ripening. Para-chlorophenoxyac$ $-100 mg/L at flowering improves fruit set under low and high temperature extremes.$

- **Canopy and crop management:** Staking tomato plants and using drip irrigation keep foliage and fruits off the ground, improve spray coverage, reduce leaf dampness, and maintain adequate canopy ventilation around reproductive structures. Ecological

engineering practices using border barrier crops (5–6 rows of sorghum or maize planted 2 months before transplanting) and intercrops (cowpea, maize, coriander, urdbean, or marigold) suppress sucking pests and protect developing flowers.

- **Pests and diseases affecting reproduction:** Gram pod borer (*Helicoverpa armigera*) larvae bore into flowers and developing fruits, controlled by spraying Novaluron 10% EC or Chlorantraniliprole 18.5% SC @ 3 ml per 10 L of water. Intercropping marigold (25-day-old seedlings with 25-day-old tomato) synchronizes flowering to trap borer moths for manual destruction of eggs and larvae on flowers. Whiteflies (*Bemisia tabaci*) disrupt flower formation and transmit Tomato leaf curl virus (ToLCV), which causes severe flower drop before fruit set. Late blight (*Phytophthora infestans*) induces floral discoloration and flower abortion. Septoria leaf spot (*Septoria lycopersici*) attacks stems and flowers. Red spider mites (*Tetranychus* spp.) impair flower and fruit formation. Root-knot nematodes (*Meloidogyne* spp.) reduce flowering and fruit production.

- **Flower drop and fruit-setting disorders:** Flower drop is caused by high temperatures (($T_{max} > 40.0°C$)), $water stress during flowering, ToLCV virus infection, and late blight floral attack. Documented fruit-setting disorders include puffiness (cavities caused by inadequate pollination/fertilization under tempera facing (blossom-end deformity from low temperatures, herbicide drift, orthrips feeding); blossom-end rot (localized Ca deficiency from soil moisture fluctuations or excess N/K); and zippering (thin longit

- **Fruit-setting assessment:** Quantitative measurement procedures, thresholds, and flower-to-fruit conversion formulas for evaluating fruit set are Not specified in the provided sources. Field assessment relies on monitoring the number of fruits per cluster, fruit retention rates, and the incidence of aborted or deformed fruits.

- **Growth monitoring and scheduling:** Monitoring protocols include regular AESA field scouting for pest-to-defender ratios, tracking high temperatures (($T_{max} > 40.0°C$)), and maintaining irrigation schedules ($5--6 days in summer, 7--8 days in kharif, 10--12 days in rabi$). Fertilizer top-dressing is scheduled at 30 and 50--60 days post-transplanting, and PG sowing.

- **Effects on crop performance:** Successful flowering and fruit setting directly govern total fruit count, average fruit weight (66.67–120.37 g), pericarp thickness (0.30–0.97 cm), total soluble solids (4.17–5.73 (°Brix)), and overall fruity ield ($20 - -24 t/ha$ in general production upto $80 - -90 t/ha$ in triple-resistant hybrids like Arka Apeksha and Arka Vishesh).

- **Integrated management workflow:** The integrated sequence begins with nursery bed preparation on broad beds with organic seed treatment; transplanting 4-week-old seedlings with 5–6 true leaves alongside barrier and trap crops; enforcing uninterrupted irrigation throughout flowering and fruit set; applying split NPK fertilizers at transplanting, 30, and 50–60 days; administering foliar PGR sprays (GA ($_3$), Triacontanol, PCPA) at 25, 45, and 65 days; implementing pest controls with yellow sticky traps and targ